Rene Hampölz
Noah Quinz
Jakob Thaller

Laborübung

23.11.2021

Halbleiterkennlinien

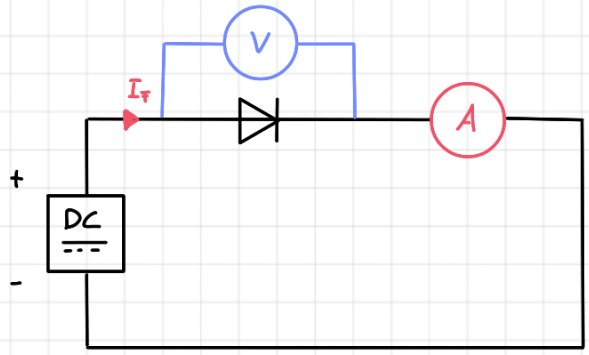
Aufgabenstellung

Kennlinien verschiedener Dioden ermitteln

Strombegrenzung: 20mA

Silizium-Diode

Schaltung

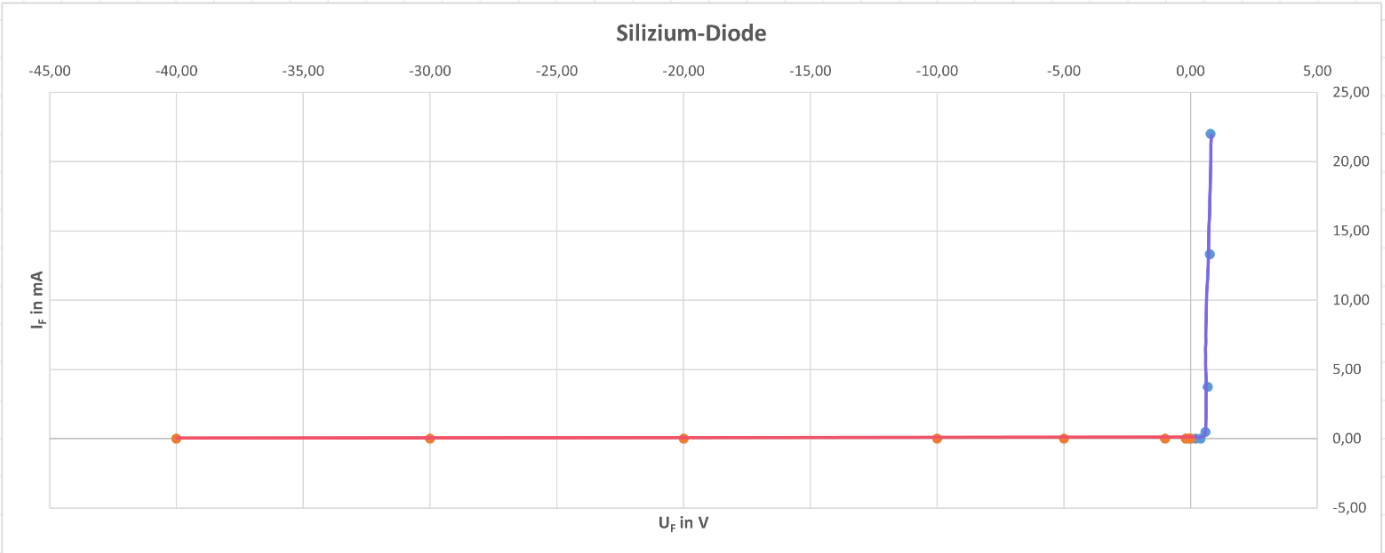


Messwerte

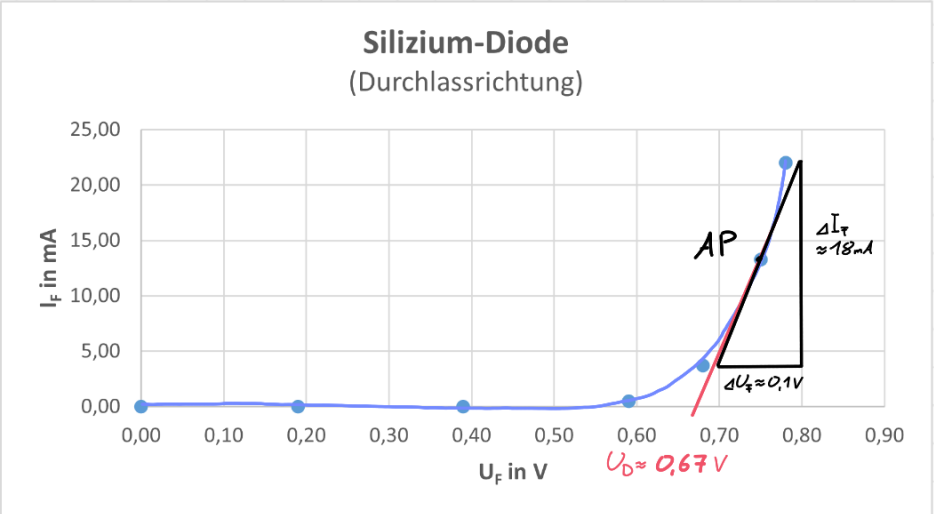
Silizium-Diode

Durchlassrichtung			*Sperrrichtung*		
$_I$ in V	U_F in V	I_F in mA	U_q in V	U_F in V	I_F in µA
0,00	0,00	0,00	0,00	0,00	0,00
0,20	0,19	0,00	-0,10	-0,10	0,00
0,40	0,39	0,00	-0,20	-0,20	0,00
0,60	0,59	0,50	-1,00	-1,00	-0,10
0,70	0,68	3,73	-5,00	-5,00	-0,50
0,80	0,75	13,30	-10,00	-10,00	-1,00
0,83	0,78	22,00	-20,00	-20,00	-2,00
			-30,00	-30,00	-3,00
			-40,00	-40,00	-4,00

Auswertung

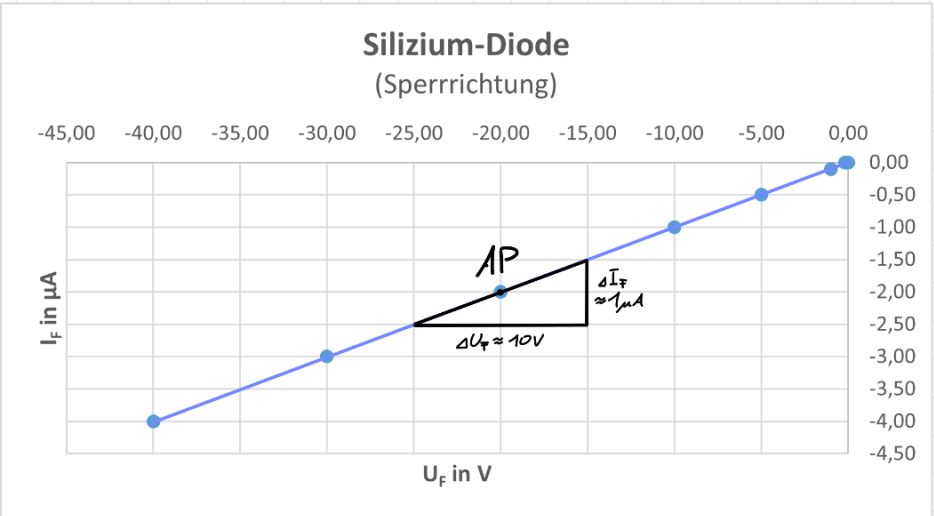


SABRINA CREATIVE DESIGN | www.stilkunst.de
DIN A4 | 5 x 5 mm | LIGHT GREY



$$r_{dif} = \frac{\Delta U_F}{\Delta I_F} = \frac{0,1}{18 \cdot 10^{-3}}$$

$$r_{dif} \approx 5,5\,\Omega$$

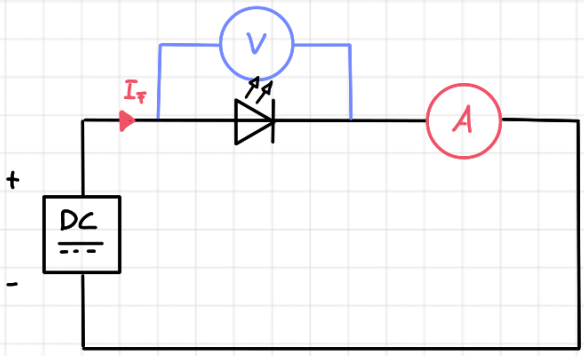


$$r_{dif} = \frac{\Delta U_F}{\Delta I_F} = \frac{10}{1 \cdot 10^{-6}}$$

$$r_{dif} \approx 10\,M\Omega$$

Leucht-Diode

Schaltung



Messwerte

Leucht-Diode

Rot			*Blau*		
$_I$ in V	U_F in V	I_F in mA	U_q in V	U_F in V	I_F in mA
0,00	0,00	0,00	0,00	0,00	0,00
0,50	0,50	0,00	0,50	0,50	0,00
1,00	1,00	0,00	1,00	1,00	0,00
1,50	1,50	0,00	1,50	1,50	0,00
1,70	1,70	0,02	2,00	2,00	0,00
1,80	1,80	0,32	2,50	2,50	0,05
1,90	1,89	2,65	2,60	2,60	0,28
2,00	1,98	6,60	2,70	2,70	0,91
2,10	2,07	11,74	2,80	2,80	2,40
2,20	2,16	16,92	2,90	2,89	5,10
2,30	2,26	23,61	3,00	2,98	9,35
			3,10	3,06	14,94
			3,20	3,14	21,80

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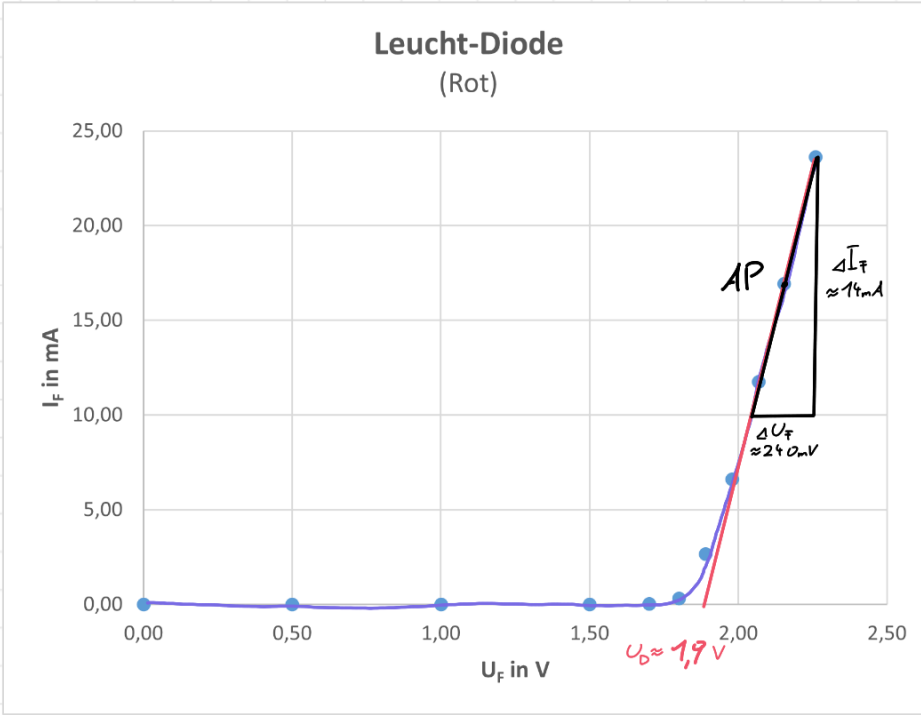
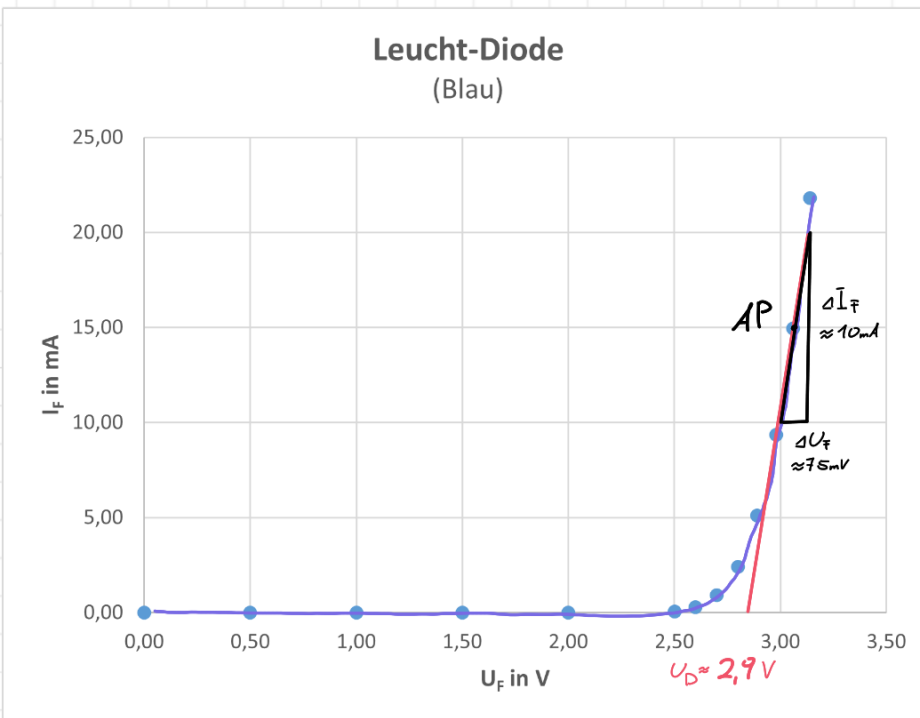
Auswertung



$$r_{dif} = \frac{\Delta U_F}{\Delta I_F} = \frac{240 \cdot 10^{-3}}{14 \cdot 10^{-3}}$$

$$r_{dif} \approx 17\,\Omega$$



$$r_{dif} = \frac{\Delta U_F}{\Delta I_F} = \frac{75 \cdot 10^{-3}}{10 \cdot 10^{-3}}$$

$$r_{dif} \approx 7,5\,\Omega$$

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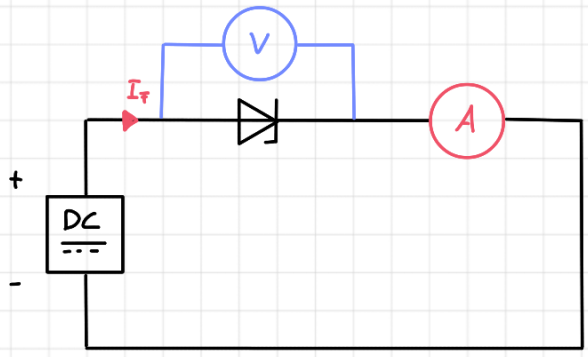
Zener-Diode

Schaltung

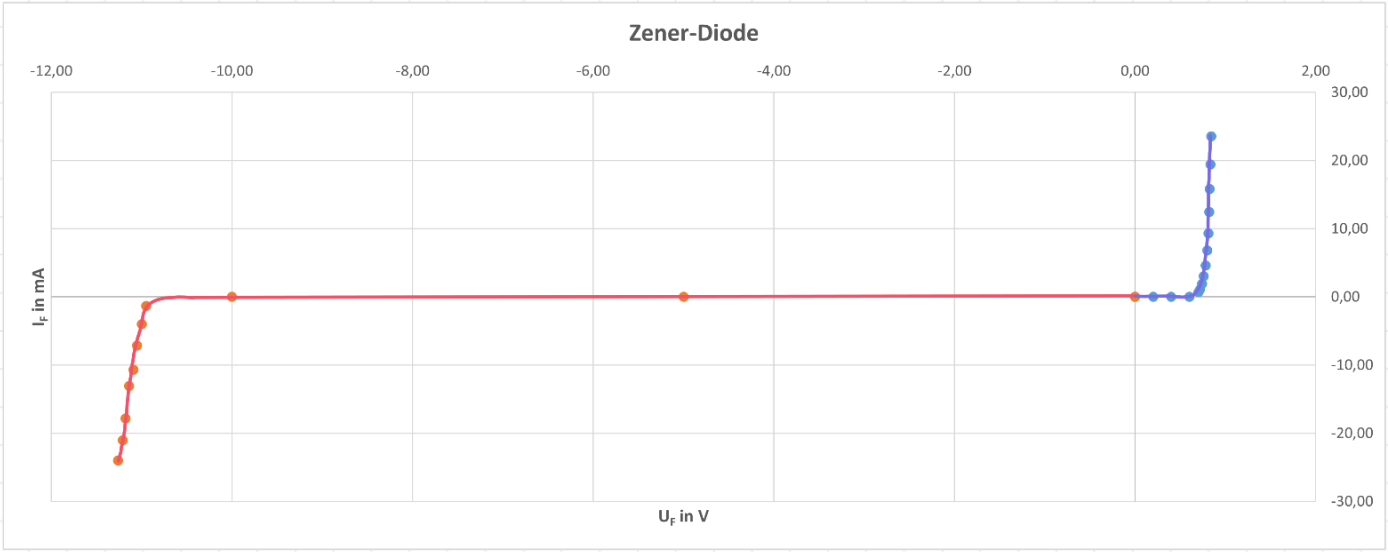


Messwerte

Zener-Diode

Durchlassrichtung			*Sperrrichtung*		
$_I$ in V	U_F in V	I_F in mA	U_q in V	U_F in V	I_F in mA
0,00	0,00	0,00	0,00	0,00	0,00
0,20	0,20	0,00	-5,00	-5,00	0,00
0,40	0,40	0,00	-10,00	-10,00	0,00
0,60	0,60	0,03	-10,95	-10,95	-1,30
0,70	0,70	0,70	-11,00	-11,00	-4,00
0,72	0,72	1,15	-11,05	-11,05	-7,10
0,74	0,74	1,92	-11,10	-11,09	-10,70
0,76	0,76	3,04	-11,15	-11,14	-13,03
0,78	0,78	4,61	-11,20	-11,18	-17,80
0,80	0,80	6,83	-11,25	-11,21	-21,00
0,82	0,81	9,34	-11,30	-11,26	-24,00
0,84	0,82	12,44			
0,86	0,83	15,84			
0,88	0,83	19,44			
0,90	0,84	23,55			

Auswertung



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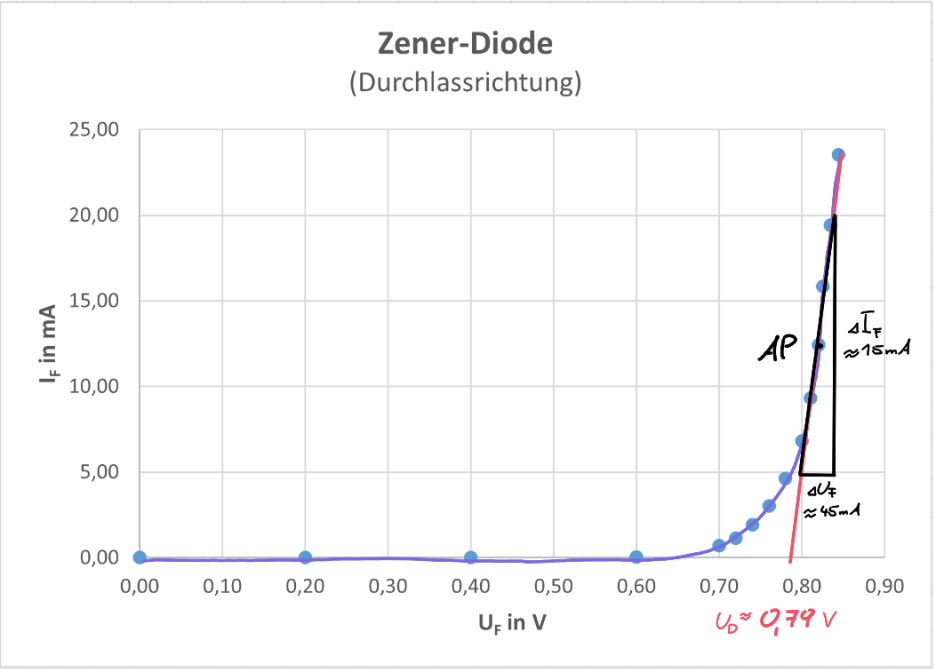
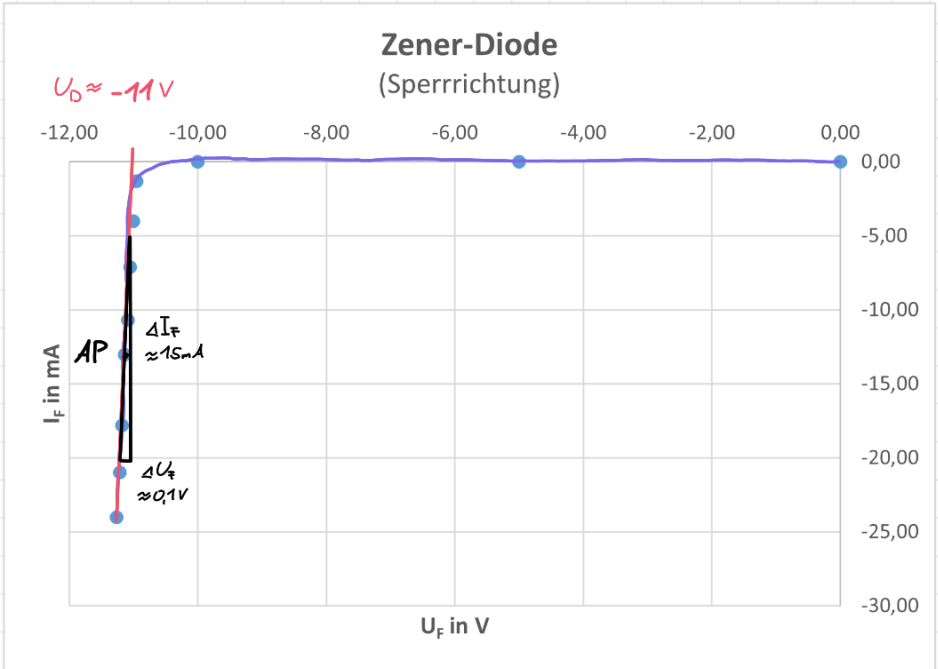


$$r_{dif} = \frac{\Delta U_F}{\Delta I_F} = \frac{45 \cdot 10^{-3}}{15 \cdot 10^{-3}}$$

$$r_{dif} \approx 3\,\Omega$$



$$r_{dif} = \frac{\Delta U_F}{\Delta I_F} = \frac{0{,}1}{15 \cdot 10^{-3}}$$

$$r_{dif} \approx 6{,}6\,\Omega$$

Verwendete Geräte

Multimeter	ET-MTL1-DM22	U_F
	ET-MAL1-DM16	I_F

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